

North Utilities - Operator Rounds and Alarm Notes

Evidence window: 2026-08-03 through 2026-08-16. These notes support alarm timing, visible leakage, unusual noise, LOTO observations, and handover context only. Instrumented records control measured values.

ID	Time	Asset	Operator note
OP-001	2026-08-03 07:08	CHWP-2	Low delta-T alarm; light mist at lower guard drain. Pump remained online. No flush gauge visible.
OP-002	2026-08-04 09:42	STR-CHW2	Basket opened by maintenance. Operator observed lint film and pea gravel. No DP indicator installed at panel.
OP-003	2026-08-05 16:10	CPL-CHWP-2	After spider work, vibration alarm returned within 25 minutes. Audible 1x hum at pump end.
OP-004	2026-08-06 08:22	TT label	BAS screen showed CT4 fan DE hot, but mechanic used handheld probe at CHWP-2 DE bearing housing.
OP-005	2026-08-07 17:20	CHWP-2 seal	Seal job marked complete. Mist still visible at guard drain during restart.
OP-006	2026-08-08 15:44	alignment	Alignment crew said site sheet passed. Vibration alarm remained after restart.
OP-007	2026-08-08 16:02	SFP-CHW2	New flush gauge read 17 psi with pump running.
OP-008	2026-08-09 14:35	STR-CHW2	First time two strainer pressure taps seen on local panel.
OP-009	2026-08-10 11:40	HX-2	Heat exchanger backflush finished, but HX DP alarm returned during evening round.
OP-010	2026-08-11 09:30	CT-4	Tower screen matted. Basin has white scale flakes and broken fill fragments.
OP-011	2026-08-12 09:52	FT-CHW2	BAS flow number changed after controls tech scaling correction. Ultrasonic meter left on pipe for comparison.
OP-012	2026-08-12 15:40	CT-4	Tower approach improved after basin screen wash. Pump still in vibration alarm.
OP-013	2026-08-15 15:45	STR-CHW2	Strainer basket pulled again, heavily blinded with fill chips and gasket fragments. Flow improved after cleaning.
OP-014	2026-08-16 06:55	Chiller 2	Short chiller safety shutdown after low-flow alarm. Tower approach normal at time of trip.

Alarm Event Register

ID	Time	Alarm	Asset	Description
ALM-001	2026-08-03 07:06	LOW-DELTAT	CHWP-2	delta-T below 8 F for 12 minutes
ALM-002	2026-08-05 16:35	PUMP-VIB	CHWP-2	drive-end vibration alarm returned after coupling spider change
ALM-003	2026-08-07 17:24	SEAL-LEAK	CHWP-2	seal leak still active after WO-92027 closeout
ALM-004	2026-08-08 15:48	PUMP-VIB	CHWP-2	alarm after site-pass alignment
ALM-005	2026-08-10 18:20	HX-DP-HIGH	HX-2	HX DP alarm after backflush
ALM-006	2026-08-12 09:48	FLOW-SCALE	FT-CHW2	operator noticed flow step after BAS correction
ALM-007	2026-08-12 10:05	TOWER-APPROACH	CT-4	approach high before basin cleaning
ALM-008	2026-08-15 14:08	STR-DP-HIGH	STR-CHW2	action-level strainer DP
ALM-009	2026-08-16 06:50	CHILLER-SAFETY	Chiller 2	six minute shutdown after low flow; reset without chiller WO

LOTO observations transcribed from shift board

Permit	Work order	Board status	Use in reconciliation
LOTO-4104	WO-92006	complete	strainer clean; no DP taps existed
LOTO-4112	WO-92027 / WO-92031	permit signed out before WO close	seal work claim remains disputed
LOTO-4118	WO-92034	missing verification signature	alignment involved pump, motor and coupling train
LOTO-4130	WO-92060	incomplete - second lock missing	HX backflush has LOTO gap
LOTO-4144	WO-92095	not attached	BV-410 trend check not stroke verified

Shift handover notes

ID	Time	Handover note
SH-001	2026-08-07 19:00	Seal mist still visible. Do not accept seal replacement without correct kit issue and leak test.
SH-002	2026-08-08 19:00	Alignment site pass did not clear vibration. Pipe strain suspected.
SH-003	2026-08-12 19:00	Tower approach improved after cleaning; pump vibration remained high.
SH-004	2026-08-15 19:00	Strainer cleaning improved flow. Retest after alignment and debris source correction.
SH-005	2026-08-16 12:00	Chiller trip treated as impact of condenser-water flow issue; no independent chiller work order requested.

Evidence boundary note

Operator notes do not control numerical pressure, flow, vibration, temperature, or inventory values when instrumented records exist. They support sequence, visible leakage, unusual noise, alarm timing, and handwritten LOTO observations only.